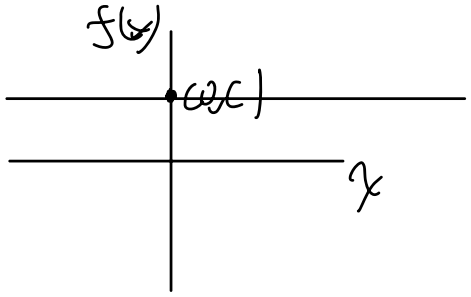
Elementary Functions

An **elementary function** is a familiar basic function whose equation, graph, domain, and range serve as useful references.

Function	Equation	Domain	Range
Constant	$f(x) = c$	$(-\infty, \infty)$	$\{c\}$
Identity	$f(x) = x$	$(-\infty, \infty)$	$(-\infty, \infty)$
Square	$f(x) = x^2$	$(-\infty, \infty)$	$[0, \infty)$
Cube	$f(x) = x^3$	$(-\infty, \infty)$	$(-\infty, \infty)$
Square root	$f(x) = \sqrt[2]{x}$	$[0, \infty)$	$[0, \infty)$
Cube root	$f(x) = \sqrt[3]{x}$	$(-\infty, \infty)$	$(-\infty, \infty)$
Reciprocal	$f(x) = \frac{1}{x}$	$(-\infty, 0) \cup (0, \infty)$	$(-\infty, 0) \cup (0, \infty)$
Absolute value	$f(x) = \lvert x \rvert$	$(-\infty, \infty)$	$[0, \infty)$
Greatest integer	$f(x) = \lfloor x \rfloor$	$(-\infty, \infty)$	$\mathbb{Z}$

- constant function

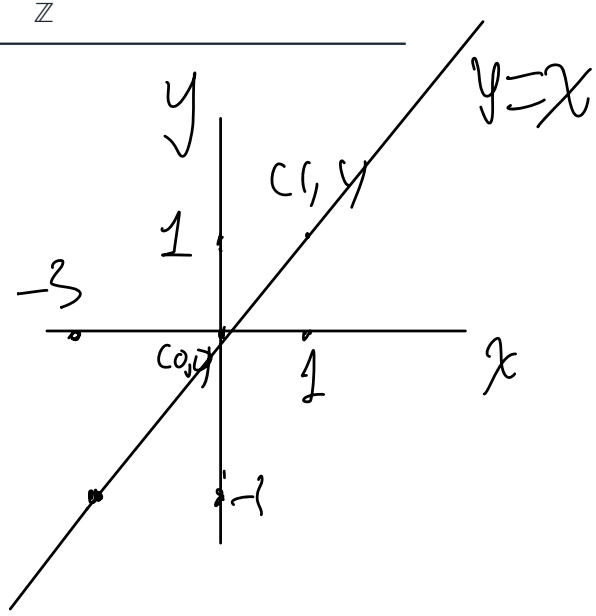
$f(x) = c$



- identity.

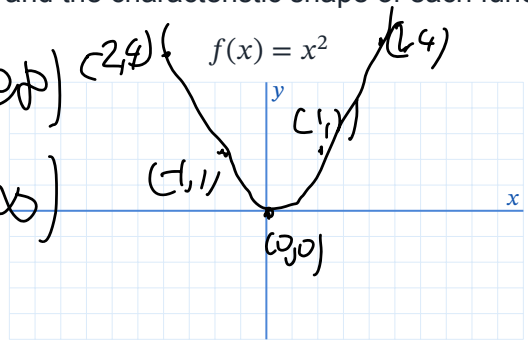
$f(x) = x$

$y = x$



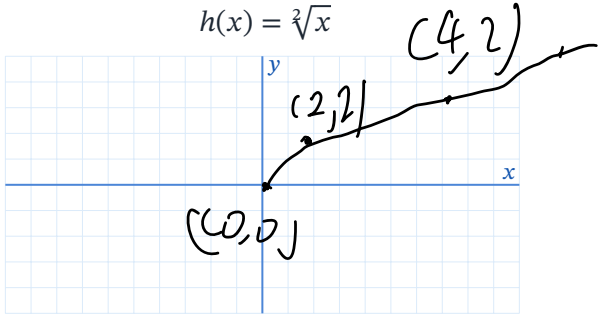
Sketch Four Elementary Functions

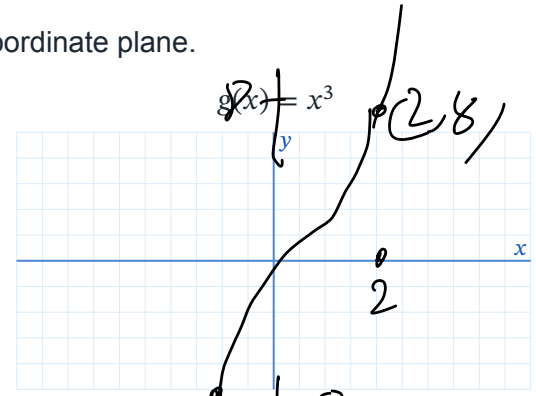
Use key points and the characteristic shape of each function. Keep each graph in its own coordinate plane.

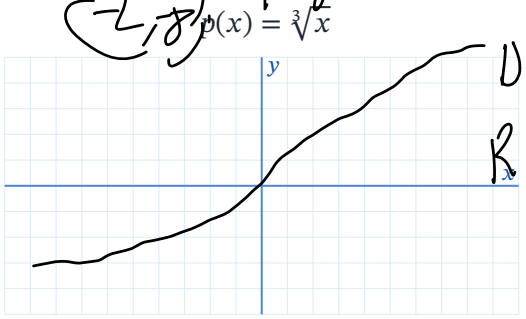


Domain: $(-\infty, \infty)$

range: $[0, \infty)$







D: $(-\infty, \infty)$

R: $(-\infty, \infty)$

- draw graph means plug in points

$f(x) = x^2$

x	$f(x)=y$
-2	4
-1	1
0	0
1	1
2	4

- $f(x) = \sqrt[2]{x}$

x	$f(x)$
0	0
1	1
4	2
9	3

Domain: $[0, \infty)$

range: $[0, \infty)$

$f(x) = x^3$

x	$f(x)$
-2	-8
-1	-1
0	0
1	1
2	8

Domain: $(-\infty, \infty)$

range: $(-\infty, \infty)$

Piecewise-Defined Functions

• $f(x) =$ formula

Definition

A function is **piecewise-defined** when different formulas are used on different intervals of the domain. Choose the formula whose condition contains the input. A closed endpoint is included; an open endpoint is excluded.

Let

$$f(x) = \begin{cases} x + 2, & x < 1, \\ x^2 - 1, & x \geq 1. \end{cases}$$

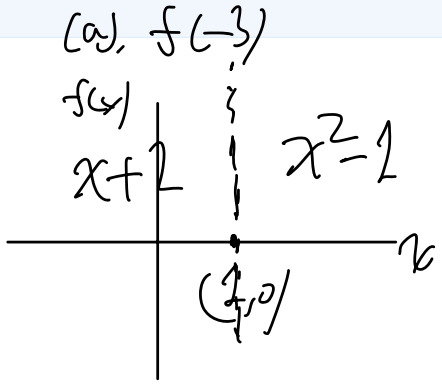
0.99+2=2.99

a. Find $f(-3)$, $f(0)$, $f(1)$, and $f(3)$.

b. For each input, explain which formula applies.

c. Predict what happens at the boundary $x = 1$. skip for now.

(a). $f(-3)$



• $-3 < 1$, so $f(-3)$ should plug in $x+2$

$f(-3) = -3+2 = -1$

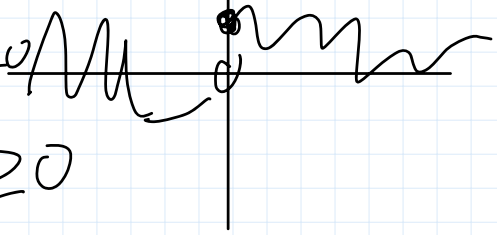
• $f(0)$, $0 < 1$, $f(0) = 0+2 = 2$

• $f(1)$, $1 = 1$, $f(1) = 1^2 - 1 = 0$

• $f(3)$, $3 > 1$, $f(3) = 3^2 - 1 = 8$

• abstract example.

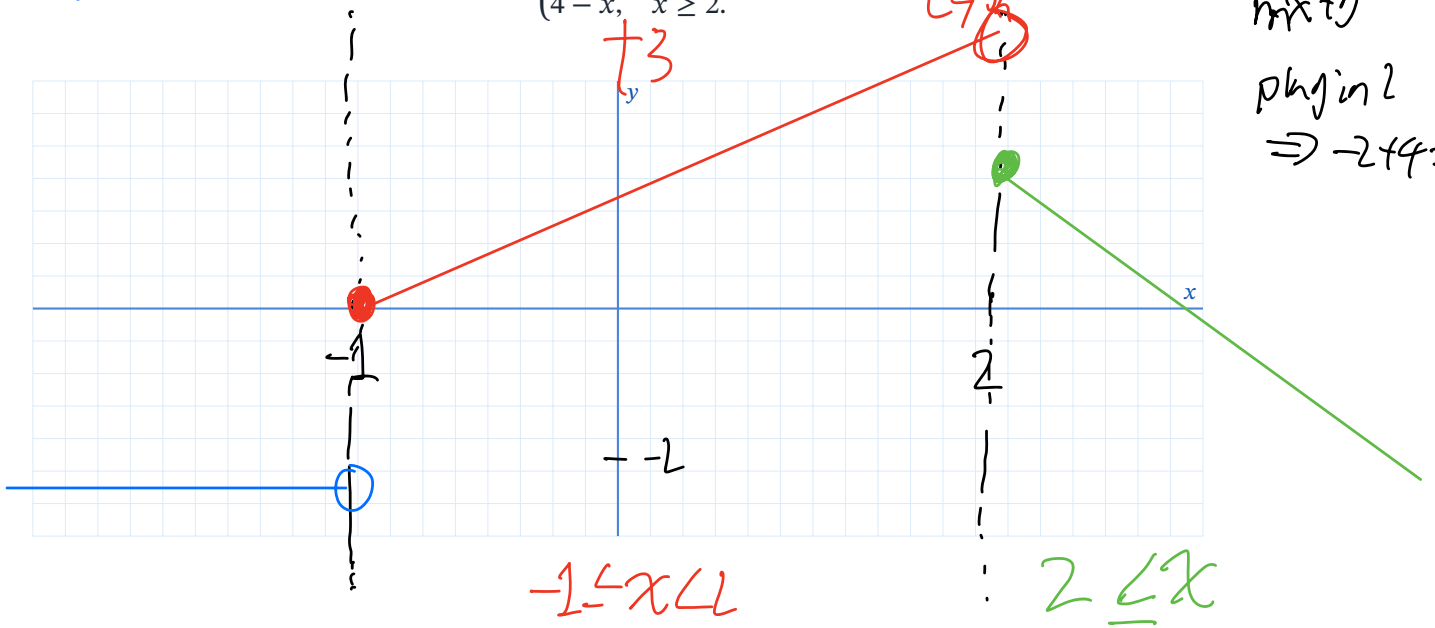
$f(x) = \begin{cases} \text{---}, & x < 0 \\ \text{---}, & x \geq 0 \end{cases}$



Graph a Piecewise Function

Graph both pieces and mark the boundary points correctly.

$$g(x) = \begin{cases} -2, & x < -1, \\ x + 1, & -1 \le x < 2, \\ 4 - x, & x \ge 2. \end{cases}$$



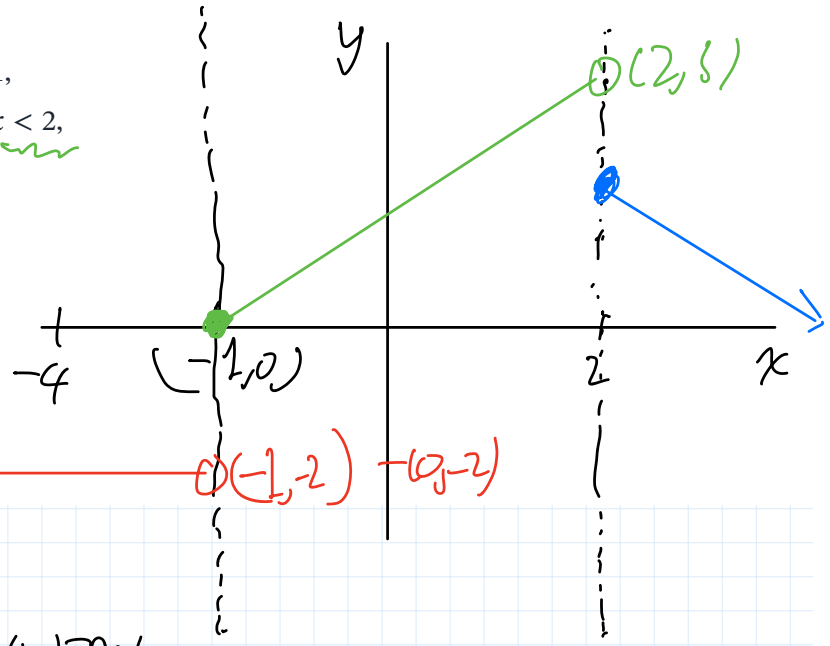
- $f(x)=mx+b$, linear function. $\begin{cases} m>0, & \nearrow \\ m<0, & \searrow \\ m=0, & \text{——} \end{cases}$

- draw graph for piecewis function, observe how many part there are. then draw a graph for each part.

Analyze a Piecewise Function

Use the function from the previous page.

$$g(x) = \begin{cases} -2, & x < -1, \\ x + 1, & -1 \le x < 2, \\ 4 - x, & x \ge 2. \end{cases}$$



a. Evaluate $g(-4)$, $g(-1)$, $g(0)$, $g(2)$, and $g(5)$.

b. Find the x- and y-intercepts.

c. State the domain and range.

d. State where g is increasing, decreasing, or constant.

(2): $g(x) = x + 1$

$g(-2) = 0$ $g(2) = 3$

$g(2) = 2$

(a) $g(-4) = -2$.

when $x = -1$, use $g(x) = x + 1$

$g(-1) = -2 + 1 = 0$

$g(0)$, use $g(x) = x + 1$

$g(0) = 0 + 1 = 1$

$g(5) = 4 - 5 = -1$

b. Find x- and y- intercept.

$g(0) = 0+1$ (using ②, $g(x)=x+1$)

$= 1$

$(0,1)$ is y-intercept.

x-intercept:

$g(x) = x+1 = 0$

$x = -1$.

$(-1, 0)$

$g(x) = 4 - x = 0$

$-x = -4$

$x = 4$

$(4, 0)$

c). Domain: $(-\infty, \infty)$

Range: $(-\infty, 3)$

Transformation Notation

Starting with an elementary function $y = f(x)$, a transformed graph can be written as

$$y = a\, f\big(b(x-h)\big) + k$$

Feature	Symbol	Effect on the graph
Vertical shift	k	$k > 0$: up; $k < 0$: down
Horizontal shift	h	$x - h$: right h units; $x + h$: left h units
Vertical scale	a	$\lvert a\rvert > 1$: stretch; $0 < \lvert a\rvert < 1$: compression
Horizontal scale	b	$\lvert b\rvert > 1$: compression by $\frac{1}{\lvert b\rvert}$; $0 < \lvert b\rvert < 1$: stretch by $\frac{1}{\lvert b\rvert}$
Reflection	sign	$a < 0$: across the x-axis; $b < 0$: across the y-axis

Reading order

First identify the elementary function. Then read horizontal changes inside $f(\,)$ and vertical changes outside $f(\,)$.

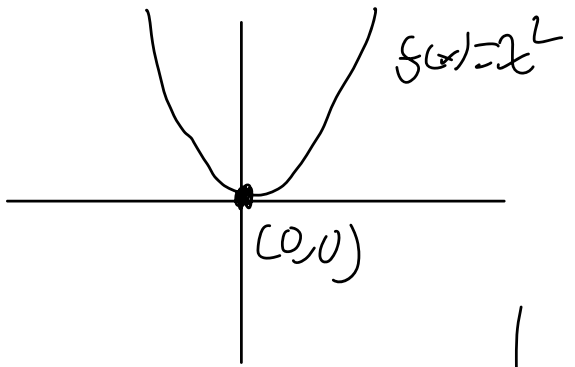
- Transformation "move" or "change" the function.
- start with elementary function.

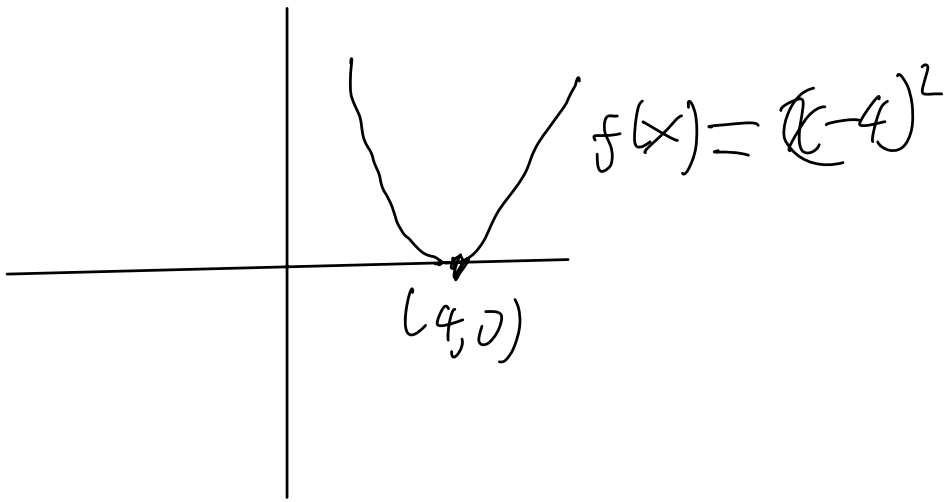
pick $f(x)=x^2$

- horizontal shift.

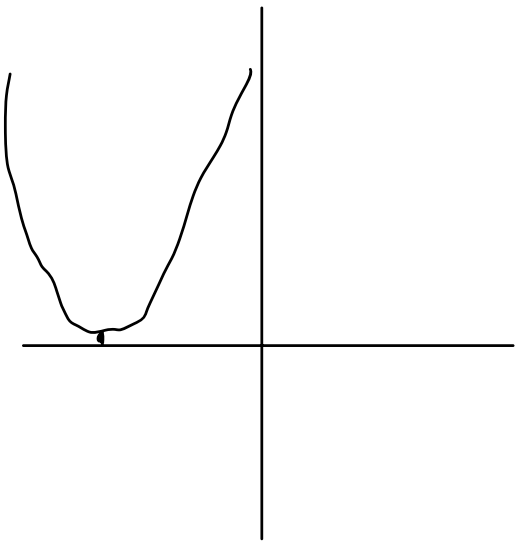
$f(x)$, if we plug in $x-h$, $\Rightarrow f(x-h)$, h is a number. pick $h=4$.

$f(x) \rightarrow f(x-4)$, this tells me $f(x)$ move right by 4 unit.





$f(x-(-4))$

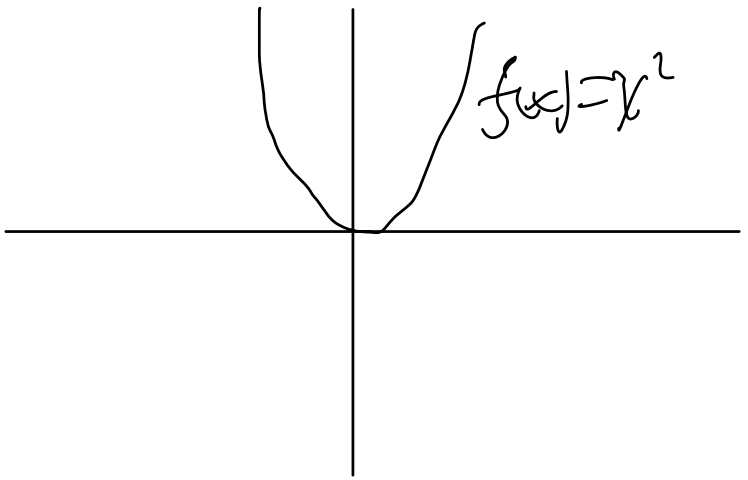


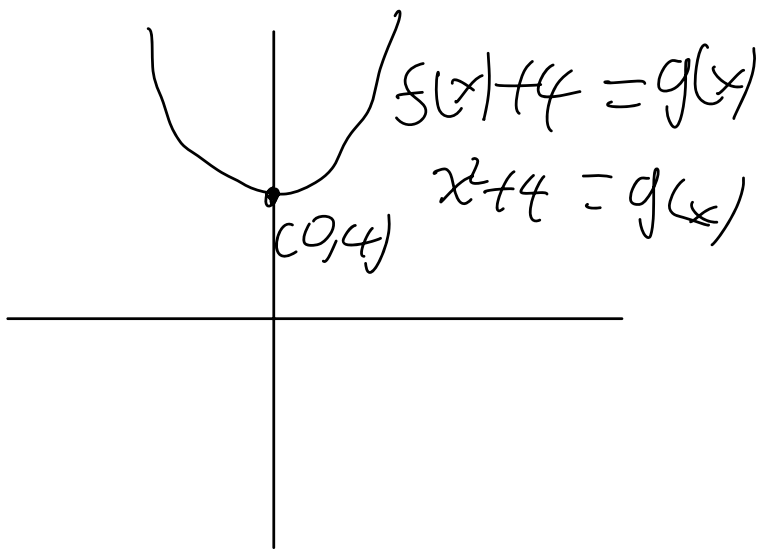
Vertical shift:

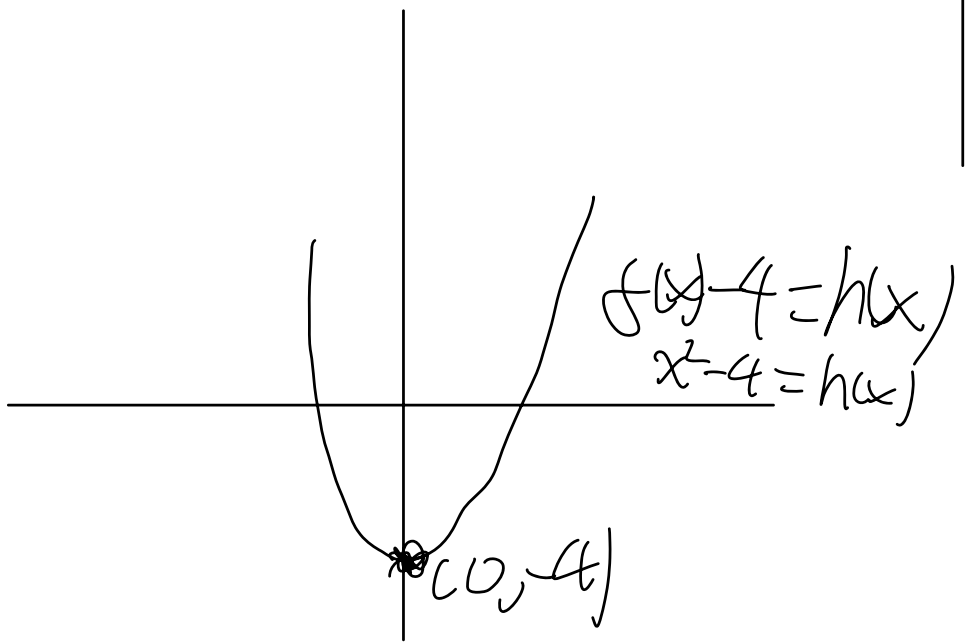
$f(x) = x^2$, $f(x) + k$, pick $k = 4$.

$= f(x) + 4$

$= x^2 + 4$, up by $k/4$ unit







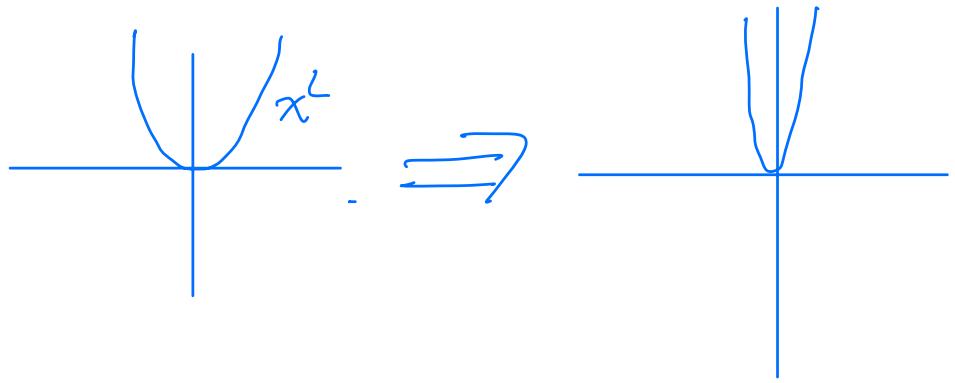
Stretch and compression:

Vertical stretch and compression:

$f(x)=x^2$, to tretch or compress $f(x)=x^2$, we have ax^2.

CaseI: is $a>1$, then it is vertical stretch.

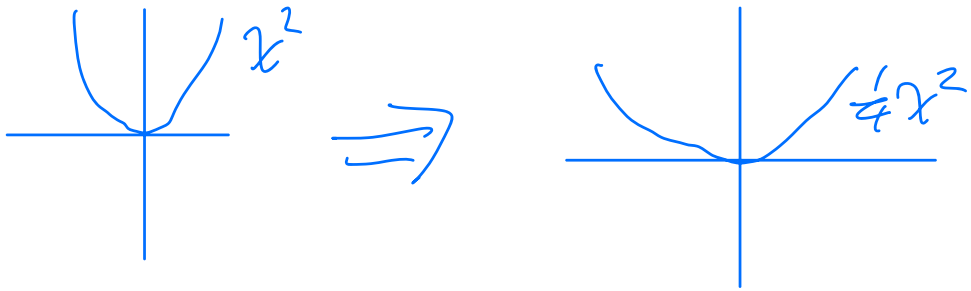
ex:



$a=4$,

CaseII: if $0<a<1$, then it is vertical compress.

ex: $a=\frac{1}{4}$,



horizental stretch and compress

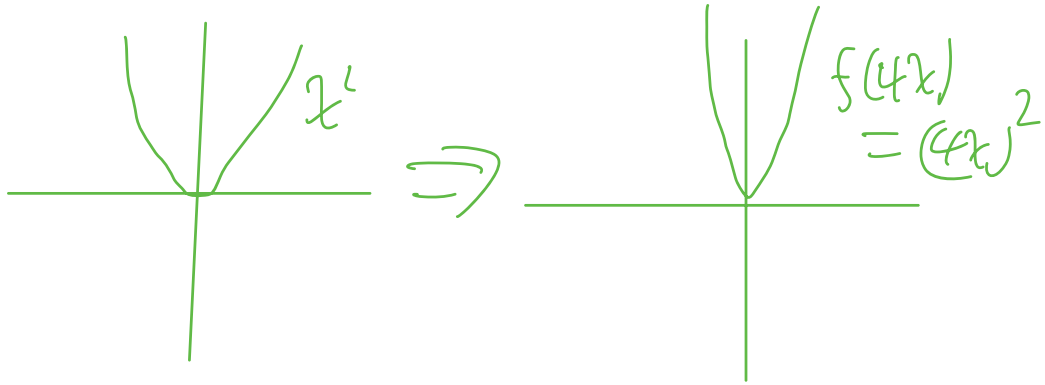
$f(x)=x^2$, is we stretch or compress horzi-cally,

we need plug bx into $f(x)$, $f(bx)$.

case I: is $b>1$, then it is horizentally compress.

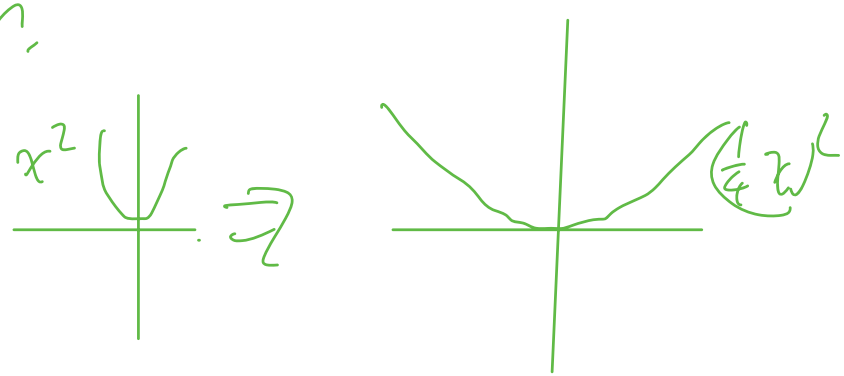
ex:

Pick $b=4$. $f(4x^2)$.



case II: a $b<1$, hoz.... stretch.

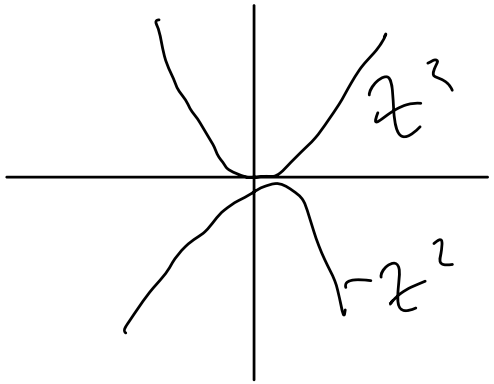
ex: $b=\frac{1}{4}$, $f(\frac{1}{4}x)=(\frac{1}{4}x)^2$



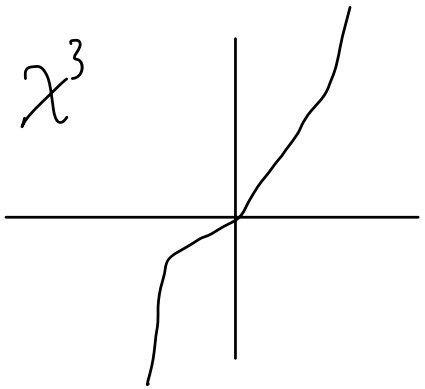
flip:

$f(x) = x^2$.

$-x^2$. flip along x-axis.

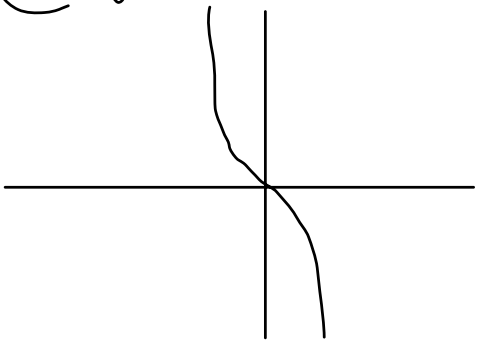


$f(x) = x^3$



$f(-x) = (-x)^3$

flip along y-axis



graph the function.

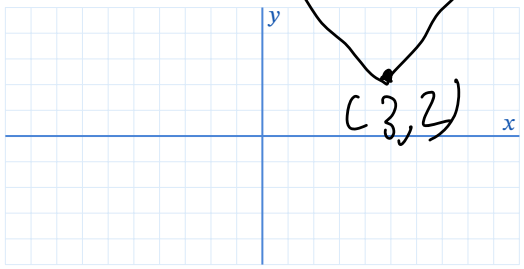
Vertical and Horizontal Shifts

Vertical shifts

$$y = f(x) + k$$

$k > 0$ moves the graph up; $k < 0$ moves it down.

$g(x) = |x - 3| + 2$

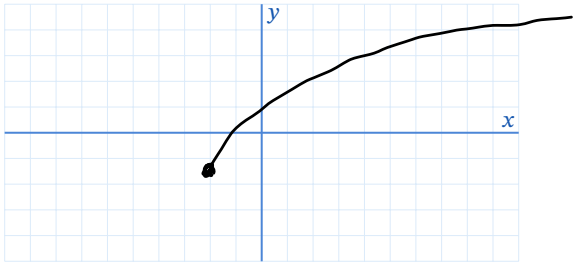


Horizontal shifts

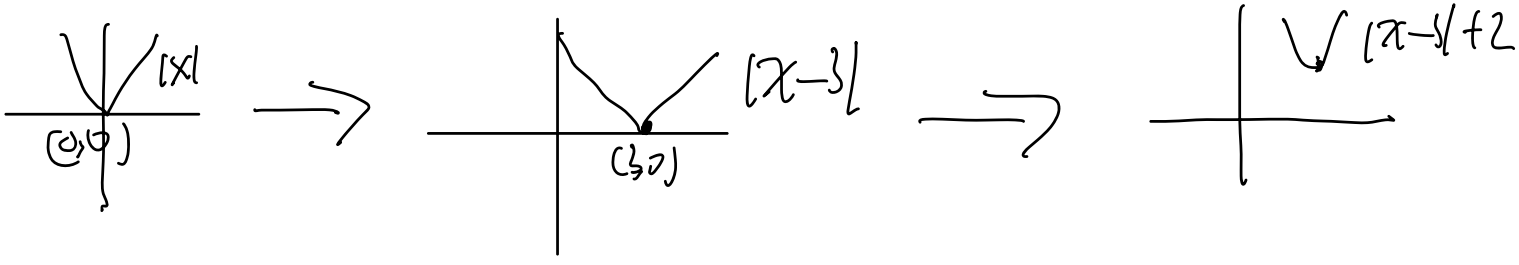
$$y = f(x - h)$$

$h > 0$ moves the graph right; replacing h by $-h$ moves it left.

$h(x) = \sqrt[3]{x + 2} - 1$

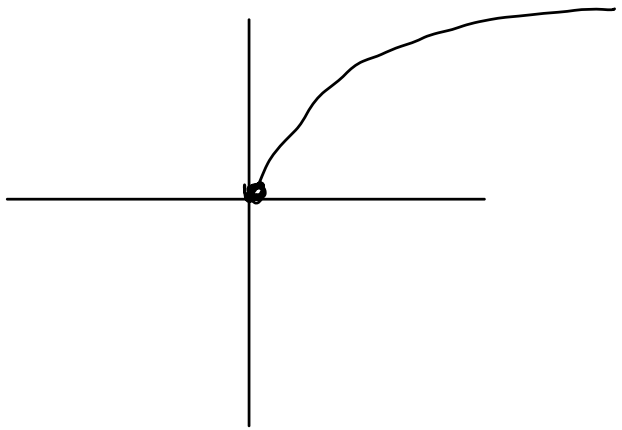


- what is the elementary/parent function?
  $|x|$ is the parent function.
- $|x| \xrightarrow[\text{to right by 3}]{\text{hor... shift}} |x-3| \xrightarrow[\text{up by 2}]{\text{vertic. shift}} |x-3|+2$



$h(x) = \sqrt[3]{x+2} - 1$

- $\sqrt[3]{x}$
- left by 2, down by 1.



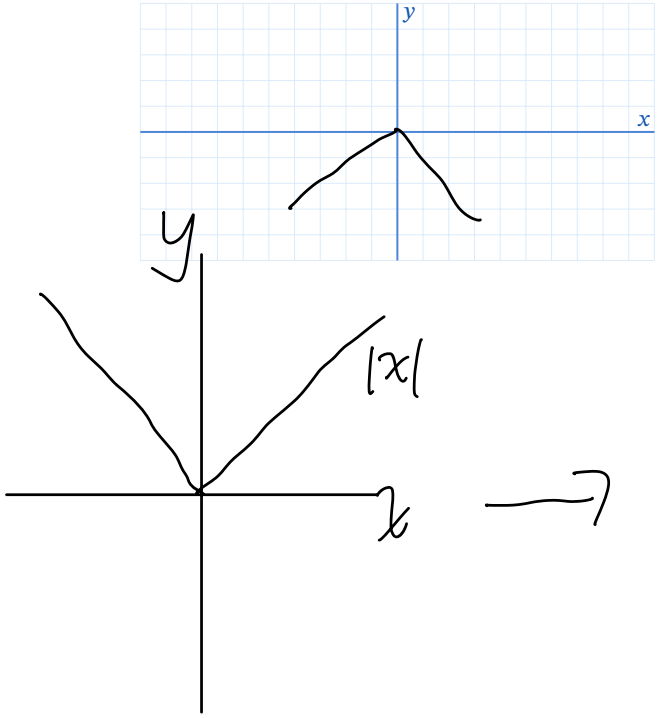
Reflections

Across the x-axis

$$y = -f(x)$$

Each output changes sign: $(x, y) \mapsto (x, -y)$.

$$g(x) = -|x|$$

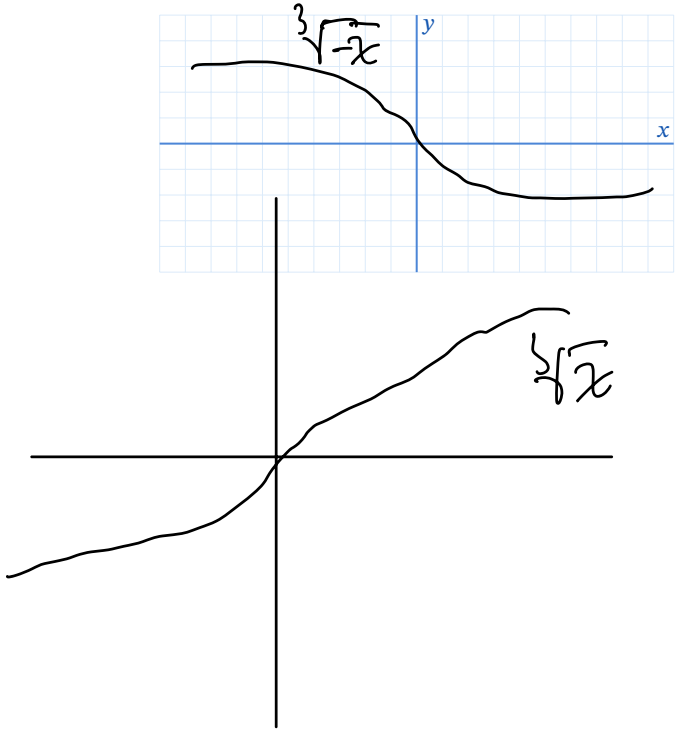


Across the y-axis

$$y = f(-x)$$

Each input changes sign: $(x, y) \mapsto (-x, y)$.

$$h(x) = \sqrt[3]{-x}$$

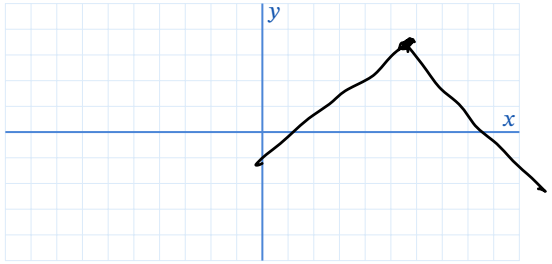


Combine Transformations

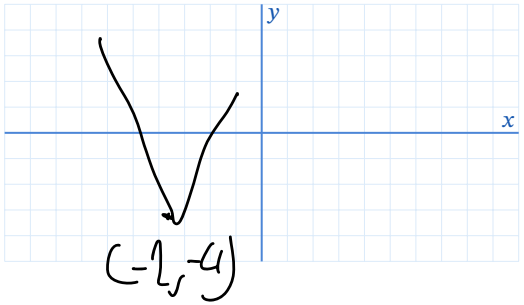
Start with $f(x) = |x|$. For each function, identify the transformations, locate the new vertex, and sketch the graph.

$$g(x) = -\frac{1}{2}|x-2|+3 \qquad h(x) = 2|x+1|-4$$

$g(x) = -\frac{1}{2}|x-2|+3$



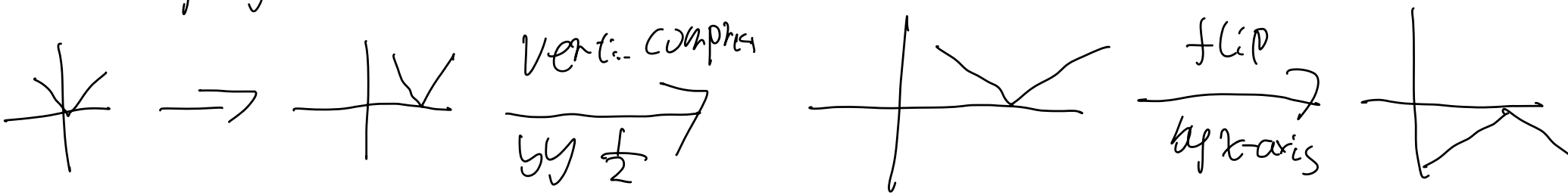
$h(x) = 2|x+1|-4$



$g(x) = -\frac{1}{2}|x-2|+3$

- $|x|$ is parent function.

$|x| \xrightarrow[\text{right by 2}]{\text{hor...}} |x-2| \longrightarrow \frac{1}{2}\cdot|x-2| \longrightarrow -\frac{1}{2}|x-2| \xrightarrow[\text{by 3}]{\text{up}} -\frac{1}{2}|x-2|+3$



Mixed Transformation Challenge

Graph the function from its elementary function.

$$y = -2\sqrt[2]{x+3}+4$$

a. Name the elementary function. $\sqrt[2]{x}$

b. List the transformations in a sensible order.

right by 3, up by 4, flip by x-axis, Vertically stretch by 2.

c. State the domain and range.

D: $x \geq -3$

R: $y \leq 4$

